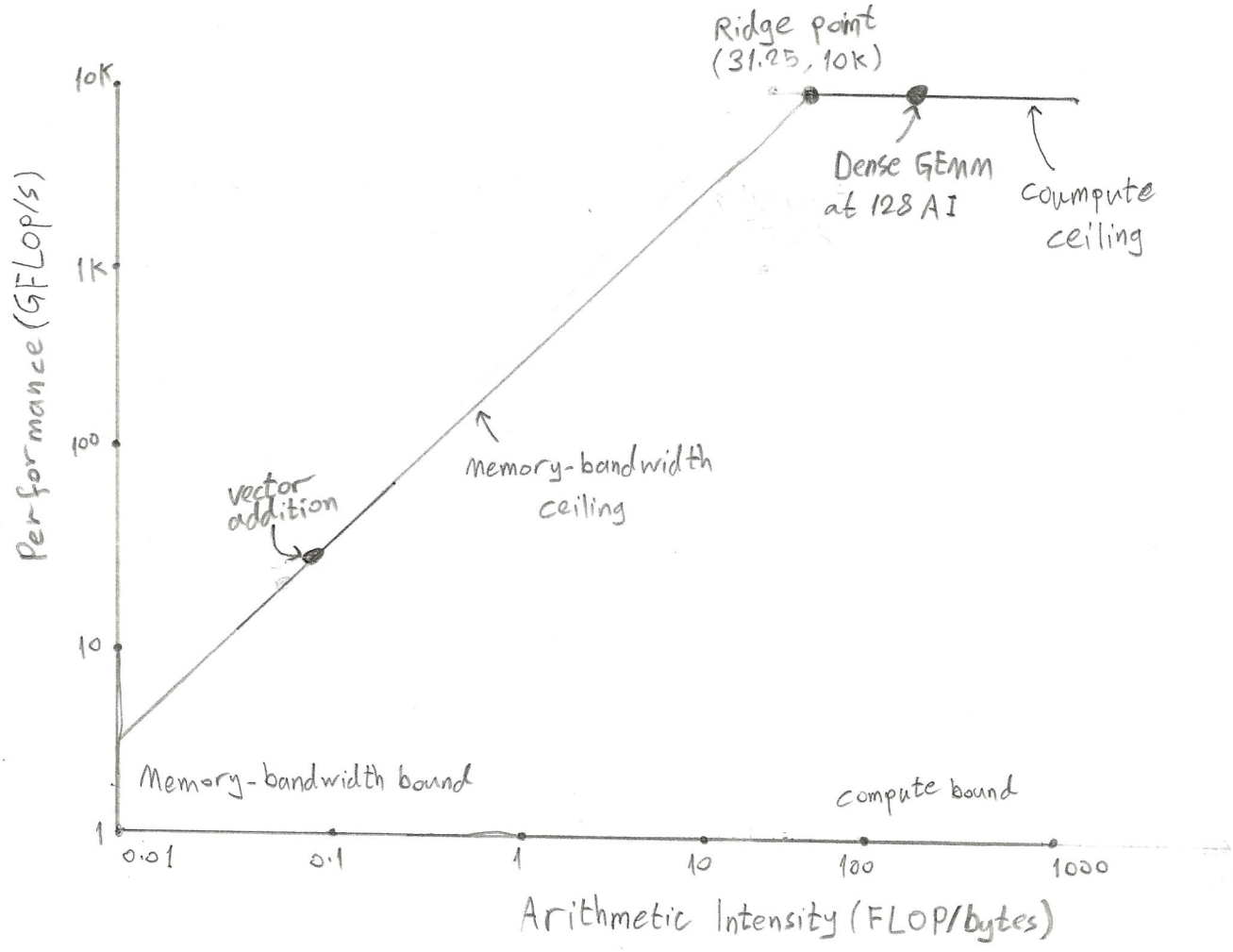

Kernel A (Dense GEMM):

FLOPs = $2N^3 = 2(1024)^3 = 2\,147\,483\,648$ FLOPs

Matrices A and B read N^2 times while matrix C read and write once

Bytes = $N^2 + N^2 + 2(N^2) = [(1024)^2 + (1024)^2 + 2(1024)^2] * 4 = 16\,777\,216$ bytes

Arithmetic intensity = FLOPs / Bytes = 2147483648 / 16 777 216 = 128 FLOPs/bytes

Kernel B (Vector addition):

FLOPs = Length = 4 194 304 FLOPs

Kernel B reads two vector inputs and write one vector output

Bytes = $[3 * 4\,194\,304] * 4 = 50\,331\,648$ bytes

Arithmetic intensity = FLOPs / Bytes = 4194304 / 50 331 648 = 0.083 FLOPs / bytes

Kernel A (Dense GEMM) is compute bound while Kernel B (vector addition) is memory-bandwidth bound. The attainable performance for Kernel A is 10K GFLOP/s while kernel B has attainable performance of (0.083 * 320 = 26.6 GFLOP/s).

Since kernel A is compute bound, adding more compute units should raise the compute ceiling and improve the performance.

Since kernel B is memory-bandwidth bound, increasing the DRAM bandwidth should increase the slope of the bandwidth and therefore improves the performance.